



Lecture 6

IP ADDRESSING



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Internet Protocol

Internet protocol properties

For the network access, you need to set these properties.

- IP address
- Subnet mask
- Default gateway(= Ethernet interface IP)
- DNS server

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: . . .

Subnet mask: . . .

Default gateway: . . .

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

☐ Validate settings upon exit

Advanced...

OK Cancel



Internet Protocol

IP configuration

It will show you window IP configuration

```
C:\C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\Wembe>ipconfig

Windows IP Configuration

Ethernet adapter 무선 네트워크 연결:

    Connection-specific DNS Suffix  . : mzumbe.ac.tz
    IP Address. . . . . : 172.30.14.221
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.30.14.1

Ethernet adapter 로컬 영역 연결:

    Connection-specific DNS Suffix  . : mzumbe.ac.tz
    IP Address. . . . . : 172.30.14.158
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.30.14.1

C:\Documents and Settings\Wembe>
```

IP Address

Internet Protocol Address (IP Address)

An Internet Protocol address (IP address) is an address used to uniquely identify a device on an IP network.

- Made up of 32 binary bits
- Be divisible into a network portion and host portion
- The 32 binary bits are broken into four octets (1 octet = 8 bits)
- Each octet is converted to decimal and separated by a period (dot)

◆ An IP address sample

00001010.00000001.00010111.00010011

**32 binary
bits**



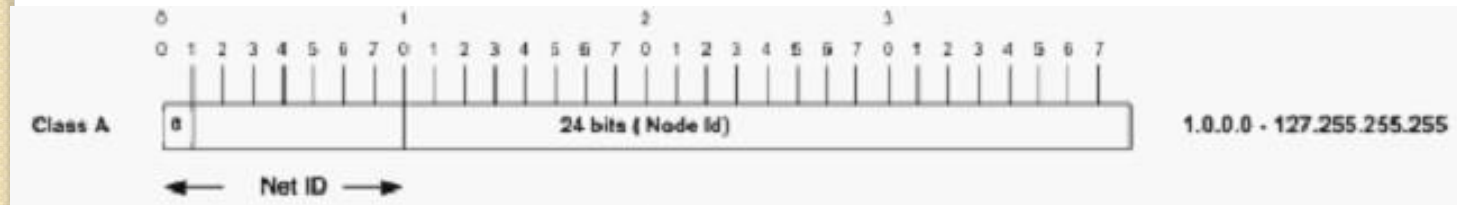
10. 1. 23. 19

Decimal address

Network Portion / Host Portion

IP Address

➤ Class A

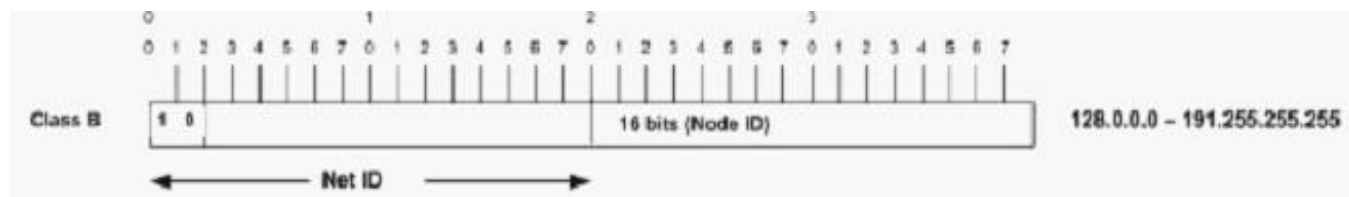


- Start with “0”
- Network portion: 1 octet
- Host portion: 3 octets
- 0.0.0.0~127.255.255.255
- No use network ID number 0 and 127

- $126(2^7-2)$ networks
- No use host ID number x.0.0.0(network IP/network prefix) and x.255.255.255(broadcast IP)
- $16,777,214(2^{24}-2)$ host IP addresses per network

IP Address

➤ Class B

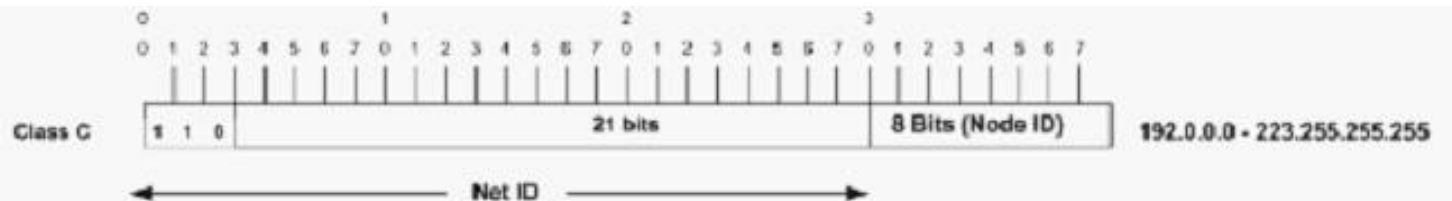


- Start with “10”
- Network portion: 2 octets
- Host portion: 2 octets

- 128.0.0.0~191.255.255.255
- No use network IP 128.0.0.0 and 191.255.0.0
- 16,384($2^{14}-2$) networks
- No use host ID number x.x.0.0(network IP/network prefix) and x.x.255.255(broadcast IP)
- 65,534($2^{16}-2$) host IP addresses per network

IP Address

➤ Class C



- Start with “110”

- Network portion: 3 octets
- Host portion: 1 octet
- 192.0.0.0~223.255.255.255
- No use network IP 192.0.0.0 and 223.255.255.0
- 2,097,150($2^{21}-2$) networks
- No use host ID number x.x.x.0(network IP/network prefix) and x.x.x.255(broadcast IP)
- 254(2^8-2) host IP addresses per network

IP Address

IP Address Classes

Class	Lead ing bits	Size of <i>network number</i> bit field	Size of <i>rest</i> bit field	Number of networks	Addresses per network	Start address	End address
Class A	0	8	24	128(2^7)	16,777,216 (2^{24})	0.0.0.0	127.255.255.255

Class B	10	16	16	16,384(2^{14})	65,536 (2^{16})	128.0.0.0	191.255.255.255
Class C	110	24	8	2,097,152(2^{21})	256 (2^8)	192.0.0.0	223.255.255.255
Class D (multicast)	1110	not defined	not defined	not defined	not defined	224.0.0.0	239.255.255.255
Class E (Experimental)	1111	not defined	not defined	not defined	not defined	240.0.0.0	255.255.255.255

SUBNET MASK

- It Identifies the two sections of the IP address, network and host fields

- Used by routers to work out the network in which a particular host belongs to
- Written either as DOTTED DECIMAL (Example 255.255.255.0) or SLASH NOTATION (Example /24)



Subnet Mask

Default subnet mask

CLASS A (1-126)

Default subnet mask = 255.0.0.0



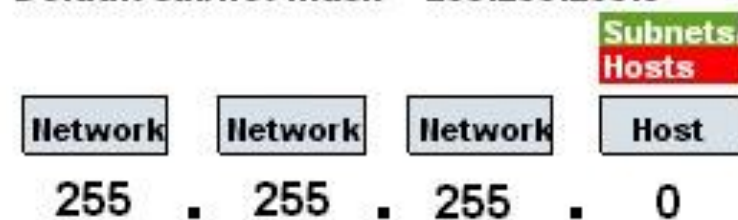
CLASS B (128-191)

Default subnet mask = 255.255.0.0



CLASS C (192-223)

Default subnet mask = 255.255.255.0



➤ **Address class A**

- Subnet mask: 255.0.0.0 • 16,777,214 network devices in one domain

➤ **Address class B**

- Subnet mask: 255.255.0.0
- 65,534 network devices in one domain

➤ **Address class C**

- Subnet mask: 255.255.255.0
- 254 network devices in one domain

SUBNETTING

- Subnetting is the practice of dividing a network into subnetworks.
- A subnetwork, or subnet, is a logically visible subdivision of an IP network.
- It is important to create subnetworks in order to use address space efficiently.
- Original assignment of classes wasted many IP addresses
 - For a example Class B can have 65,534 hosts in one network (This is way more than most companies require)

- Also, Class C can have 254 hosts in a single network, while a company requires smaller number of addresses.
- Therefore Subnetting allows to create smaller networks with only required number of host addresses needed

SUBNETTING Continues....

Subnetworking

- ☐ Solution is to subdivide networks
- ☐ Take a network number
- ☐ Divide it into smaller networks
- ☐ These are called subnetworks (subnets)

- To do subnetting you just need to borrow few bits of the host part and make them network part.
- You only leave bits for host part that are enough for the number of hosts per network you require
- At this moment we will only focus on class C addresses, where we deal with only the last octet in subnetting

SUBNETTING Continues....

Given a network prefix 201. 222. 5. 0 , Create 5 subnets that contains at least 20 usable IP addresses in each.

Conditions

- Network prefix: 201. 222. 5. 0
- Number of hosts needed per subnet: 20

- Number of subnets needed: 5
- Default subnet mask: 255.255.255.0 (Because the given Address fall under Class C)

👉 Solution

- ❖ From the table below, add from right until you get a number that will fit the hosts (Usable IP Addresses) you desire; Now I need at least 20 hosts

128 64 32 | 16 8 4 2 1

(Therefore; You get 31 by keeping 5 bits for Host, Thus other 3 bits will become Network Part)

New Subnet mask 255.255.255.224(11111111.11111111.11111111.11100000)

Subnet	Network Prefix	Usable IP Addresses (Host IP)	Broadcast
1 st Subnet	201.222.5.0	201.222.5.1 – 201.222.5.30	201.222.5.31
2 nd Subnet	201.222.5.32	201.222.5.33 – 201.222.5.62	201.222.5.63
3 rd Subnet	201.222.5.64	201.222.5.65 – 201.222.5.94	201.222.5.95
4 th Subnet	201.222.5.96	201.222.5.97 – 201.222.5.126	201.222.5.127
5 th Subnet	201.222.5.128	201.222.5.129 – 201.222.5.158	201.222.5.159

SUBNETTING Continues....

Given a network prefix 192.168.1. 0 , Create 3 subnets that contains at least 50 usable IP addresses in each.

Conditions

- Network prefix: **192.168.1. 0**
- Number of hosts needed per subnet: 50
- Number of subnets needed: 3
- Default subnet mask: 255.255.255.0 (Because the given Address fall under Class C)

👉 Solution

- ❖ From the table below, add from right until you get a number that will fit the hosts (Usable IP Addresses) you desire; Now I need at least 50 hosts

128 64 | 32 16 8 4 2 1

(Therefore; You get 63 by keeping 6 bits for Host, Thus other 2 bits will become Network Part)

New Subnet mask 255.255.255.192(11111111.11111111.11111111.11000000)

...

Subnet	Network Prefix	Usable IP Addresses (Host IP)	Broadcast
1 st Subnet	192.168.1.0	192.168.1.1 - 192.168.1.62	192.168.1.63
2 nd Subnet	192.168.1.64	192.168.1.65 - 192.168.1.126	192.168.1.127
3 rd Subnet	192.168.1.128	192.168.1.129 - 192.168.1.190	192.168.1.191

SUBNETTING Continues....

Given a network prefix 192.168.1.0 , Create 2 subnets that contains at least 14 usable IP addresses in each.

Conditions

- Network prefix: **192.168.1.0**
- Number of hosts needed per subnet: 14
- Number of subnets needed: 2
- Default subnet mask: 255.255.255.0 (Because the given Address fall under Class C)

👉 Solution

- ❖ From the table below, add from right until you get a number that will fit the hosts (Usable IP Addresses) you desire; Now I need at least 14 hosts

128 64 32 16 | 8 4 2 1

(Therefore; You get 15 by keeping 4 bits for Host, Thus other 4 bits will become Network Part)

New Subnet mask 255.255.255.240(11111111.11111111.11111111.11110000)

...

Subnet	Network Prefix	Usable IP Addresses (Host IP)	Broadcast
1 st Subnet	192.168.1. 0	192.168.1. 1 - 192.168.1. 14	192.168.1. 15
2 nd Subnet	192.168.1. 16	192.168.1. 17 - 192.168.1. 30	192.168.1. 31